

Foster (Yun-Fei) TENG

University of Toronto | foster.teng@mail.utoronto.ca | <https://flyincloud2001.github.io/My-Webpage/>

RESEARCH INTERESTS

Mathematical Physics, Partial Differential Equations, Algebra, General Relativity, Quantum Mechanics

EDUCATION

University of Toronto (cGPA: 3.82/4.0) Fall 2022 - Present
Honours Bachelor of Science (HBSoc), Physics and Mathematics Specialist

HONORS & AWARDS

Margaret Ronald Taylor and Thomas Paxton Taylor Scholarships in Mathematics - CAD 3,055 2024-2025
NSERC Undergraduate Student Research Award (USRA) - CAD 7,500 Summer 2025
The Class of 3T0 and Associates Scholarship in Mathematics and Physics - CAD 1,910 2023-2024
University of Toronto Dean's List 2023, 2024, 2025

RESEARCH EXPERIENCE

Visualization of Algebraic Structures(Advisor: Professor Camelia Karimianpour) Summer 2025
Visualization of finite cyclic groups and their products, Cayley tables, Cayley diagrams, using MATLAB
Geodesic Trajectories on Hyperbolic Space(Advisor: Professor Wenyu Pan) Winter 2025
Hyperbolic Geometry, Horocycles and Busemann Cocycles, Fuchsian Groups and Dirichlet Domains

PROFESSIONAL PROJECTS

Maple Whisper (Anonymous Chat Platform) Winter 2025
Full-stack web application with real-time chat, user matching system, and PostgreSQL database, deployed on Railway

ACADEMIC PROJECTS

The Least Action Principle for the Einstein Field Equation(Course Instructor: Professor Robert McCann)
Derivation of the Einstein Field Equation using variational calculus, including proving Palatini identity, Jacobi's formula, etc. Winter 2025
Functions Grapher
In Course Python Project, programmed by Matplotlib Fall 2022
MTA Data Analytics Case Competition
Sports Analytics, Python, CSV, & API, and Data Visualization using PowerBi Fall 2022

TALKS

Visualization of Algebraic Structures
Summer Undergraduate Research Seminar, University of Toronto Summer 2025

WORK EXPERIENCE

Videographer and Classroom Support Specialist
Record lectures, and troubleshoot computing or equipment issues in classrooms in UofT Winter 2023

Private Mathematics and science Tutor

for student in courses MAT137Y1, MAT246H1, MAT135H1

VOLUNTEER

University Events Voter

Attend regular advisory committee meeting and vote for important events in UofT

Fall 2022

Environmental Protection Volunteer Team

Trash picking and sorting around the campus during the course breaks

2019

Volunteer Care for the Handicapped

Aiwei Mercy Home for Disabled, the job was to assist the severely disabled to eat

2019